

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)**UNIT TEST****16-08-2020****JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)****ANSWER KEY****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	B	A	B,D	A,B	A,B,C,D	A,B,C	A,D	A,C
	Q.	11	12								
	A.	A,D	A,B,C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	600.00 to 604.00	13.92 to 13.93	24166 to 24167	1.00 to 1.04	6.00	64.00				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	C	D	A,C	A,B,C	B,D	A,B,C	B,C,D	B,C
	Q.	11	12								
	A.	A,B,C	A,C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	200.00	690.00	0.16 TO 0.17	361.50	4.00	6.00				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	B	B	A,B,D	B,C,D	B,D	C,D	C,D	B,D
	Q.	11	12								
	A.	B,C,D	B,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	0.25	1.73	0.25	-4.42 or -4.43	3.46	12.00				



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PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (A)

Sol. $\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{100}{1}$

$V_p = 100 V_s = 12000$

2. Ans. (C)

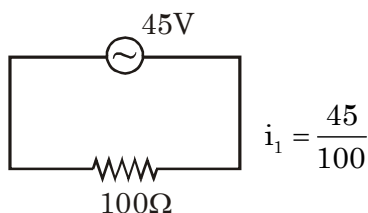
Sol. $\Delta f = f_1 - f_2$

In standing wave pattern, nodes and antinodes are formed.

3. Ans. (B)

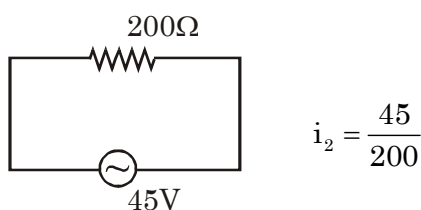
Sol. When f is small

$X_C \left(= \frac{1}{\omega C} \right)$ is very large and $X_L (= \omega L)$ is very small



when f is large

X_C is very small and X_L is very large



4. Ans. (A)

Sol. $\phi = \frac{\mu_0 i}{2\pi x} \pi r^2$

$r = \frac{\mu_0 i}{2x^2} r^2 \frac{dx}{dt}$

$= \frac{\mu_0 i}{2 \times (0.5)^2} \times (0.5)^2 \times 10^{-4} \times \sqrt{10}$

$= 2\pi \times 10^{-10} \times \pi$

$= 2 \times 10^{-9} \text{V}$

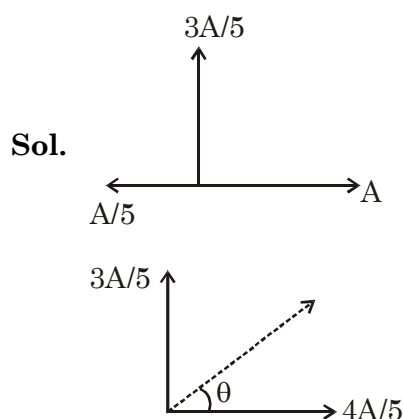
5. Ans. (B,D)

Sol. $v = \omega r$

$2\pi r = n\lambda$

$\Rightarrow f = \frac{v}{\lambda} = \frac{\omega n}{2\pi}$

6. Ans. (A,B)



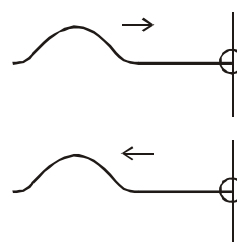
Sol.

wave is travelling +ve x, so
 $y = A \sin(\omega t - kx + \theta)$

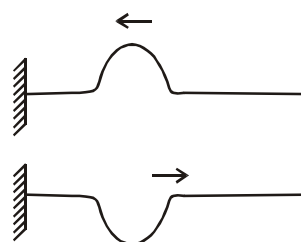
7. Ans. (A,B,C,D)

Sol. $V = \sqrt{\frac{T}{\mu}} = 10 \text{ m/s}$

on reflection from free end



on reflection from fixed end (phase changes by π)



8. Ans. (A,B,C)

Sol. $V_p = \frac{\partial y}{\partial t}$

$$P = -T \frac{\partial y}{\partial x} \frac{\partial y}{\partial t}$$

9. Ans. (A,D)

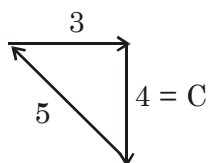
Sol. $vp_1 - vS_0 \cos(\omega t - kx)$
 $\delta p_1 = BS_0 k \cos(\omega t - kx)$
 $vp_2 = vS_0 \cos(\omega t + kx)$
 $p_2 = -BS_0 k \cos(\omega t + kx)$

10. Ans. (A,C)

Sol. $f_A = \frac{8}{8-3} \times 10 = 16 \text{ Hz}$

$$f_C = \frac{2}{2+3} \times 10 = 4 \text{ Hz}$$

$$f_B = \frac{4}{4-3 \cos 90^\circ} \times 10 = 10 \text{ Hz}$$



11. Ans. (A,D)

Sol. $C = C_v + \frac{R}{1-x}$

$$C_v = \frac{n_1 C_{v1} + n_2 C_{v2}}{n_1 + n_2}$$

$$= \frac{\frac{3R}{2} + \frac{5R}{2}}{2} = 2R$$

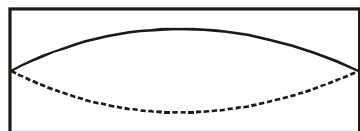
$$R = 2R + \frac{R}{1-x}$$

$$x = 2$$

$$PV^2 = \text{constant} \Rightarrow TV = \text{constant}$$

$$\Rightarrow V \propto x$$

$$v = \sqrt{\frac{\gamma RT}{M}} \quad v \propto \frac{1}{\sqrt{x}}$$



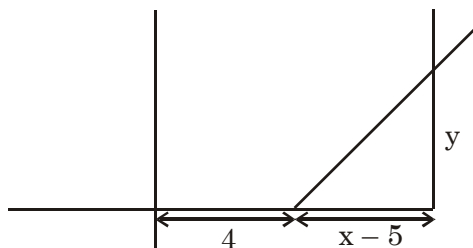
$$\frac{n\lambda}{2} = L = \frac{\text{Volume}}{\text{Area}}$$

$$\frac{n\lambda}{2} = \frac{V}{A}$$

$$\frac{nv}{2f} = \frac{V}{A} \Rightarrow n = \frac{nVf}{vA}$$

12. Ans. (A,B,C)

Sol.



$$\sqrt{y^2 + (x+5)^2} = \sqrt{y^2 + (x-5)^2} + 8$$

$$y^2 + (x+5)^2 = y^2 + (x-5)^2 + 64$$

$$20x + 64 = 16\sqrt{y^2 + (x-5)^2}$$

$$25x^2 + 256 - 160x = 16y^2$$

$$9x^2 = 16y^2 + 144$$

SECTION-II

1. Ans. 600 to 604

Sol. $V = \frac{\sqrt{T}}{\mu} \quad \left(\mu = \frac{m}{\ell} \right)$

$$P = \frac{1}{2} \mu v \omega^2 A^2$$

2. Ans. 13.92 to 13.93

Sol. $F \cos 53^\circ = 10$

$$F = \frac{50}{3}, \quad T = \frac{5}{\sqrt{\frac{50}{3m}} \times 5} = \sqrt{\frac{3m}{10}}$$

$$\frac{4F'}{5} = 10 \Rightarrow F' = \frac{50}{4}$$

$$T + 1 = \frac{8}{\sqrt{\frac{50}{4m}} \times 5} = \sqrt{\frac{4m}{10}}$$

$$\text{then } \sqrt{\frac{4m}{10}} - \sqrt{\frac{3m}{10}} = 1$$

$$\Rightarrow (2 - \sqrt{3})x = 1$$

$$\Rightarrow \sqrt{\frac{m}{10}} = x = \frac{1}{2 - \sqrt{3}}$$

$$\Rightarrow m = \frac{10}{7 - 4\sqrt{3}} \times 10^{-4}$$

3. **Ans. 241.66 to 241.67**

$$\text{Sol. } \omega_R = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10 \times 10^{-3} \times 100 \times 10^{-6}}} = 1000$$

$$\omega = 2\omega_R = 2000$$

$$X_L = \omega L = 20$$

$$X_C = \frac{1}{\omega C} = 5$$

$$P = V_{Rms} \left(\frac{V_{Rms}}{Z} \right) \left(\frac{R}{Z} \right)$$

$$Z = \sqrt{(X_L - X_C)^2 + R^2} = \sqrt{325}$$

$$\text{Energy} = P \times \text{time} = P \left(\frac{2\pi}{\omega} \right)$$

4. **Ans. 1.00 to 1.04**

$$\text{Sol. } f_1 = \frac{1}{2(\ell + 1.2r_1)} \times 320$$

$$f_2 = \frac{1}{2(\ell + 1.2r_2)} \times 320$$

$$2 = 160 \left[\frac{1}{1 + 1.2r_1} - \frac{1}{1 + 1.2r_2} \right]$$

$$\frac{1}{80} \approx 1.2(r_2 - r_1)$$

$$r_2 - r_1 = 1/96 \text{ m} = 1.00 \text{ cm}$$

5. **Ans. 6.00**

$$\text{Sol. } y_1 = 4\sin(500 \pi t) \quad y_2 = 2\sin(506 \pi t)$$

$$\text{Number of beats} = \frac{n_1 - n_2}{2} = \frac{506 - 500}{2}$$

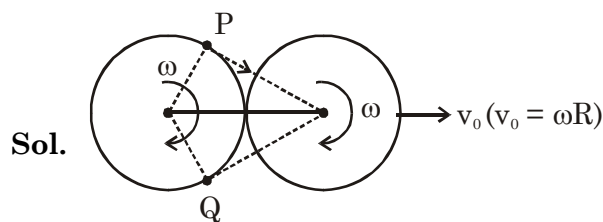
$$= 3 \text{ beats/sec}$$

$$\text{As } I_1 \propto (16) \text{ and } I_2 \propto 4$$

$$\Rightarrow \frac{I_{\max}}{I_{\min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2}$$

$$\Rightarrow \left(\frac{4+2}{4-2} \right)^2 = \left(\frac{6}{2} \right)^2 = 9$$

6. **Ans. 64.00**



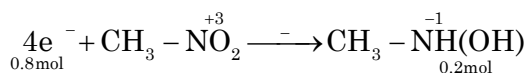
$$f_{\max} = f \left(\frac{v}{v - \omega R} \right)$$

$$f_{\min} = f \left(\frac{v}{v + \omega R} \right)$$

$$\therefore \text{bandwidth} = f_{\max} - f_{\min}$$

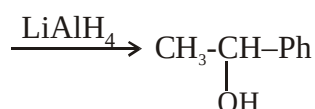
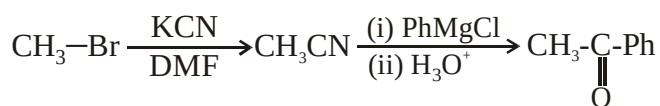
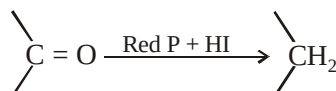
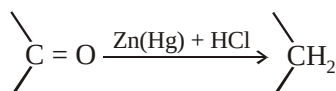
$$= f \left(\frac{v}{v - v_0} \right) - f \left(\frac{v}{v + v_0} \right)$$

$$= \frac{2fv_0 V}{v^2 - v_0^2}$$

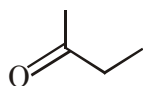
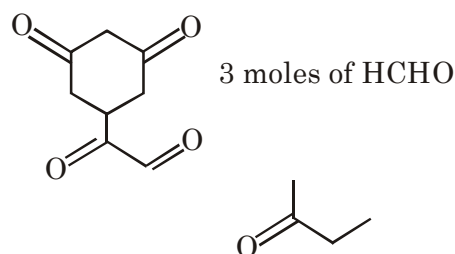
PART-2 : CHEMISTRY**SOLUTION****SECTION-I**1. **Ans.(D)**2. **Ans.(C)**

$$[A^{+2}] = \frac{K_{sp}}{[\text{OH}^-]^2} = \frac{4 \times 10^{-18}}{(10^{-5})^2} = 4 \times 10^{-8} \text{ M}$$

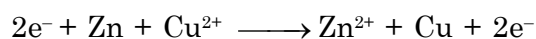
$$[B^{+2}] = \frac{0.1}{2} \text{ M}$$

3. **Ans.(C)**4. **Ans. (D)**5. **Ans. (A,C)**6. **Ans. (A,B,C)**

Sol. Ozonolysis of D option compound gives following structure

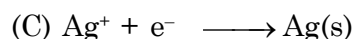
7. **Ans. (B,D)**

Sol. P₁ and P₂ is acetone and propanal. So, P₁ and P₂ can not distinguish using 2,4-DNP and NaHSO₃.

8. **Ans.(A,B,C)**9. **Ans.(B,C,D)**10. **Ans.(B,C)**11. **Ans.(A,B,C)**12. **Ans.(A,C)**

$$\begin{array}{ccc} 0.90 & 0.10 & \\ 0.50 & 0.50 & E = E^0 \end{array}$$

$$n_e = 0.8$$



$$W_{\text{ag}} = 0.8 \times 108 = 86.4 \text{ gm.}$$

SECTION-II1. **Ans.(200.00)**

$$B = -0.105 \text{ l}$$

$$C = b^2 = 4 \times 10^{-4} \Rightarrow 2 \times 10^{-2} \frac{\text{l}}{\text{mol}}$$

$$b - \frac{a}{RT} = -0.105$$

$$0.02 - \frac{a}{0.08 \times 400} = -0.105 \Rightarrow a = 4 \frac{\text{atm} - \text{l}^2}{\text{mol}^2}$$

$$\frac{a}{b} = \frac{4}{0.02} = 200$$

2. **Ans.(690.00)**

$$W = -6 \text{ bar m}^3 = -600 \text{ kJ.}$$

$$\begin{aligned} \Delta U_m &= 0.225 [P_2 V_2 - P_1 V_1] \\ &= 0.225 \times 40 = 9 \text{ kJ/mole} \end{aligned}$$

$$q = \Delta U - w$$

$$= 9 \times 10 - (-600)$$

$$= 690 \text{ kJ.}$$

3. **Ans.(0.16 to 0.17)**4. **Ans.(361.50)**5. **Ans. (4.00)****Sol.** (i) Tollens reagent

(ii) Baeyer villiger oxidation

(iii) Strong oxidising agent

(v) Strong oxidising agent

6. **Ans.(6.00)****Sol.** (a) Rosenmund reduction

(b) Stephens reaction

(d) Imine hydrolysis

(e) Dibal H reaction with acid chloride to aldehyde

(f) Cu⁺/573 K reaction of primary alcohol to aldehyde

(h) Dry distillation.

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A)**

Sol. Putting $x = \tan\theta$ and $y = \tan\phi$.

Then equation becomes

$$\sec\theta + \sec\phi = A (\tan\theta \sec\phi - \tan\phi \sec\theta).$$

$$\Rightarrow \cos\phi + \cos\theta = A(\sin\theta - \sin\phi)$$

$$\Rightarrow 2\cos\frac{\theta+\phi}{2}\cos\frac{\theta-\phi}{2} = 2A\cos\frac{\theta+\phi}{2}\sin\frac{\theta-\phi}{2}$$

$$\Rightarrow \cot\frac{\theta+\phi}{2} = A \Rightarrow \theta - \phi = 2\cot^{-1}A$$

$$\Rightarrow \tan^{-1}x - \tan^{-1}y = 2\cot^{-1}A$$

$$\text{Differentiating, we get } \frac{1}{1+x^2} - \frac{1}{1+y^2} \frac{dy}{dx} = 0$$

Which is a differential equation of degree 1.

2. **Ans. (A)**

Sol. Clearly, $\frac{P_a}{h_a} + \frac{P_b}{h_b} + \frac{P_c}{h_c}$

$$= \frac{\text{ar}(\Delta PBC)}{\text{ar}(\Delta ABC)} + \frac{\text{ar}(\Delta PCA)}{\text{ar}(\Delta ABC)} + \frac{\text{ar}(\Delta PAB)}{\text{ar}(\Delta ABC)}$$

$$= \frac{\text{ar}(\Delta PBC) + \text{ar}(\Delta PCA) + \text{ar}(\Delta PAB)}{\text{ar}(\Delta ABC)}$$

$$= \frac{\text{ar}(\Delta ABC)}{\text{ar}(\Delta ABC)} = 1$$

3. **Ans. (B)**

Sol. $(A+B)^2 = (A+B)(A+B)$

$$= A^2 + AB + BA + B^2$$

$$= A + B \text{ if } AB + BA = O$$

$$\Rightarrow AAB + ABA = O$$

$$\Rightarrow AB - BAA = O$$

$$\Rightarrow AB - BA = O$$

$$\Rightarrow AB = O \text{ \& } BA = O$$

4. **Ans. (B)**

$$\text{Sol. Limit} = \lim_{x \rightarrow 0} \frac{(\cos 2x - 1)(\cos x - e^{-x})}{-\sin x \cdot x^n}$$

$$= \lim_{x \rightarrow 0} \frac{(1 - \cos 2x)(\cos x - e^{-x})}{(2x)^2 \frac{\sin x}{x} \cdot x^{n+1}} \cdot (2n)^2$$

$$= \lim_{x \rightarrow 0} \frac{2(\cos x - e^{-x})}{x^{n-1}}$$

$$= \lim_{x \rightarrow 0} \frac{2 \left[\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} \dots \right) - \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} \dots \right) \right]}{x^{n-1}}$$

$$= \lim_{x \rightarrow 0} \frac{2[x - x^2 \dots]}{x^{n-1}} = \lim_{x \rightarrow 0} \frac{2[1 - x \dots]}{x^{n-1}} = 2$$

if $n = 2$.

5. **Ans. (A,B,D)**

$$\text{Sol. } f'(x) = \frac{3^x}{1+x^2} > 0 \quad \forall x > 0$$

$$f''(x) > 0 \quad \forall x > 0$$

So $f(x)$ and $f'(x)$ both are increasing function

$$f(x) = \int_1^x \frac{3^t}{1+t^2} dt > \int_1^x \frac{dt}{1+t^2}$$

$$\Rightarrow f(x) > (\tan^{-1}t)_1^x$$

$$\Rightarrow f(x) > \tan^{-1}x - \frac{\pi}{4}$$

$$\Rightarrow f(x) + \frac{\pi}{4} > \tan^{-1}x$$

6. **Ans. (B,C,D)**

$$\text{Sol. } f''(x) \geq -2 \quad \forall x \in [-3, 3]$$

$$\Rightarrow \int_{-3}^3 f''(x) dx \geq -2 \int_{-3}^3 1 dx \Rightarrow f'(3) - f'(-3) \geq -12$$

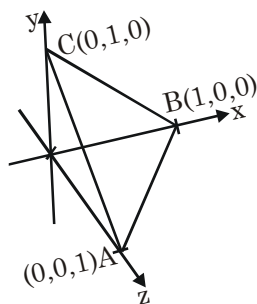
$$\Rightarrow -12 \geq -12 \Rightarrow f''(x) = -2 \quad \forall x \in [-3, 3]$$

$$\Rightarrow f(x) = -x^2 + bx + c$$

$$\therefore f(x) = -x^2 + 6x - 4$$

7. Ans. (B,D)

Sol.



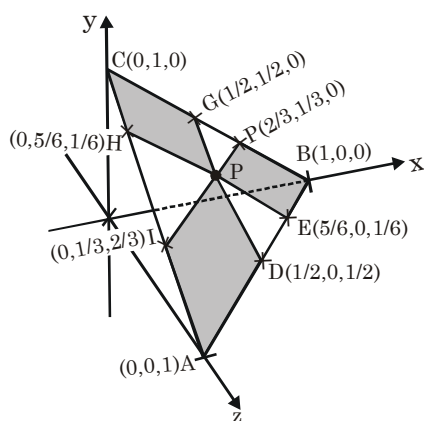
$A(\alpha)$ will be area of $\triangle ABC$ which is an equilateral

$$\Delta A(\alpha) = \frac{\sqrt{3}}{4} (\sqrt{1+1})^2 = \frac{\sqrt{3}}{2} \text{ sq. unit}$$

for $A(\beta)$, consider the point $P\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right)$ in

the plane $x + y + z = 1$

The region β is shown as shaded region



$\triangle PFG$, $\triangle PDE$ and $\triangle PHI$ are equilateral triangles and we find required area by removing their total area from $\text{ar}(\triangle ABC)$
Now, FI is line through P parallel to AB

$$\text{Equation of FI : } \frac{x - \frac{1}{2}}{1} = \frac{y - \frac{1}{3}}{0} = \frac{z - \frac{1}{6}}{-1}$$

$$\text{at } z = 0 : F\left(\frac{2}{3}, \frac{1}{3}, 0\right)$$

$$\text{at } x = 0 : I\left(0, \frac{1}{3}, \frac{2}{3}\right)$$

similarly we find points G, D and H, E

$$\therefore \text{Required Area} = \frac{\sqrt{3}}{2} - \left[\frac{\sqrt{3}}{18} + \frac{\sqrt{3}}{72} + \frac{\sqrt{3}}{8} \right]$$

$$= \frac{11\sqrt{3}}{36} \text{ sq. units}$$

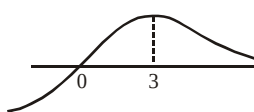
8. Ans. (C,D)

$$\text{Sol. } f'(x) = \frac{3}{2} x^{\frac{1}{2}} e^{-\frac{x}{2}} + x^{\frac{3}{2}} e^{-\frac{x}{2}} \left(-\frac{1}{2} \right)$$

$$= \frac{1}{2} x^{\frac{1}{2}} e^{-\frac{x}{2}} (3 - x)$$

$$\therefore f(x) \uparrow \text{ in } (-\infty, 3) \text{ \& \downarrow in } (3, \infty)$$

$$\therefore f(3) = \frac{3\sqrt{3}}{e\sqrt{e}}$$



$$\therefore \int_0^3 f(x) dx < 3f(3) = \frac{9\sqrt{3}}{e\sqrt{e}}$$

9. Ans. (C,D)

Sol. $\text{adj}(A) = \text{adj}(B)$

$$\Rightarrow |A|^{n-1} = |B|^{n-1}$$

$$\Rightarrow \begin{cases} |A| = |B| \text{ for even order} \\ |A| = \pm |B| \text{ for odd order} \end{cases}$$

$$\text{Now, } A^{-1}|A| = B^{-1}|B|$$

take increase on both the sides

$$\Rightarrow \frac{A}{|A|} = \frac{B}{|B|} \Rightarrow \begin{cases} A = B \text{ for even order} \\ A = \pm B \text{ for odd order} \end{cases}$$

10. Ans. (B,D)

Sol. (A) $\text{trace}(A^2 + 2AB - 4BA - 4B^2)$

$$\neq \text{trace}(A^2) - 4\text{trace}(B^2)$$

(B) $(\text{Adj } A - I)^2 - I$

$$= \text{Adj}(A^2) - 2\text{Adj } A$$

$$= \text{Adj } A (\text{Adj } A - 2I)$$

(D) $((AB^2)^2)' = (AB^2 AB^2)'$

$$= B^2 AB^2 A$$

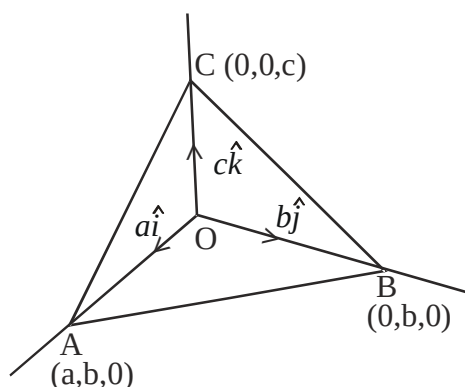
$$= (B^2 A)^2$$

11. Ans. (B,C,D)

Sol. Let equation plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

passing through (1, 1, 1)

$$\text{so, } \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$$



Volume of tetrahedron OABC

$$= \frac{1}{6} [a\hat{i} \ b\hat{j} \ c\hat{k}]$$

$$= \frac{1}{6} abc$$

because $a, b, c > 0$, so by G.M. $\geq$ H.M.

$$\sqrt[3]{abc} \geq \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}$$

$$abc \geq 27$$

$$\frac{1}{6} abc \geq \frac{9}{2}$$

$$V \geq \frac{9}{2}$$

12. Ans. (B,D)

Sol. Interchanging x & y

$$x - x^2 = mx$$

$$\Rightarrow x = 0, 1 - m$$

$$\int_0^{1-m} (x - x^2 - mx) dx = \pm \frac{9}{2}$$

$$\Rightarrow m = -2, 4$$

SECTION-II

1. Ans. 0.25

$$\text{Sol. } I = \int \frac{\cos^4 x \cdot \cos x}{\sin x} dx$$

$$\text{put } \sin x = t$$

$$= \int \frac{(1-t^2)^2 dt}{t} = \int \frac{(1+t^4-2t^2)}{t} dt$$

$$= \log|t| + \frac{t^4}{4} - t^2 + C$$

$$= \log|\sin x| + \frac{1}{4} \sin^4 x - \sin^2 x + C$$

2. Ans. 1.73

$$\text{Sol. } V_{DABC} = \frac{1}{6} = \frac{1}{3} \times (\text{area of base}) \times \text{height}$$

$$\text{Also } \frac{1}{6} \leq \frac{1}{3} \times \frac{1}{2} \times AC \times BC \times \sin 45^\circ \times AD$$

$$\therefore \left(\frac{AC}{\sqrt{2}} \right) \cdot BC \cdot AD \geq 1$$

$$\text{Now, } \frac{AC}{\sqrt{2}} + BC + AD \geq 3 \cdot \sqrt[3]{\frac{AC}{\sqrt{2}} \cdot BC \cdot AD}$$

(using A.M.- G.M.)

$$\therefore \frac{AC}{\sqrt{2}} \cdot BC \cdot AD \leq 1$$

$\therefore$ Equality must hold

$$\frac{AC}{\sqrt{2}} = BC = AD = 1$$

$\therefore$ AD must be perpendicular

$$\therefore CD^2 = AD^2 + AC^2 = 3$$

3. Ans. 0.25

$$\begin{aligned} \text{Sol. } I &= \int_0^{\pi/4} \left(\frac{\sin^2 7\theta - \sin^2 5\theta}{\sin 2\theta} + \frac{\sin^2 5\theta - \sin^2 3\theta}{\sin 2\theta} \right. \\ &\quad \left. + \frac{\sin^2 3\theta - \sin^2 \theta}{\sin 2\theta} + \frac{\sin^2 \theta}{\sin 2\theta} \right) d\theta \\ &= \int_0^{\pi/4} \left[\sin 12\theta + \sin 8\theta + \sin 4\theta + \frac{1}{2} \tan \theta \right] d\theta \\ &= - \left[\frac{\cos 12\theta}{12} + \frac{\cos 8\theta}{8} + \frac{\cos 4\theta}{4} \right]_0^{\pi/4} + \frac{1}{2} [\log |\sec \theta|]_0^{\pi/4} \\ &= - \left[\frac{-1-1}{12} + \frac{1-1}{8} + \frac{-1-1}{4} \right] + \frac{1}{2} (\log \sqrt{2}) \\ &= \frac{1}{6} + \frac{1}{2} + \frac{1}{4} \log 2 = \frac{2}{3} + \frac{1}{4} \log 2 \end{aligned}$$

4. Ans. -4.42 or -4.43

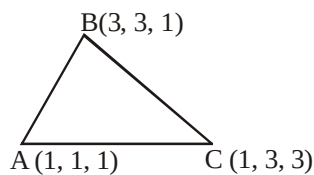
Sol. $P \equiv (1 + 2K, -K, 2K)$ in the two planes
 $1 + 2K + K + 2K - 2 = 0$

$$\Rightarrow K = 1/5. \text{ So } P \equiv \left(\frac{7}{5}, \frac{-1}{5}, \frac{2}{5} \right)$$

$$\text{Also, } \lambda \left(\frac{7}{5} \right) + 3 \left(\frac{2}{5} \right) + 5 = 0 \Rightarrow \lambda = -\frac{31}{7}$$

5. Ans. 3.46

Sol.



And $\triangle ABC$ is an equilateral triangle with
 side $= 2\sqrt{2}$

$$\therefore \text{Area} = \frac{\sqrt{3}}{4} \times 8 = 2\sqrt{3}$$

6. Ans. 12.00

Sol. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $n(A)$ denotes number of
 such matrices satisfying given conditions.

$$\text{Now, } \text{adj}A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix}$$

$$\Rightarrow d = -a$$

$$\text{Now, } |A| = ad - bc = -1$$

$$\Rightarrow a^2 + bc = 1$$

$$\text{Case I : } a = 0$$

$$\text{so, } d = 0 \text{ and } bc = 1$$

$$\Rightarrow n(A) = 2$$

$$\text{Case II : } a = 1 \text{ or } -1$$

$$\text{so, } d = -1 \text{ or } 1, \text{ and } bc = 0$$

$$n(A) = 2 \times 5$$

$$\text{hence, } n(A) = 12$$